



Storyboard for My First VST Design

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Intr o



- My program will be a VST (Virtual Studio Tech/Virtual Instrument) that a musician can use to implement different types of instruments, samples, and sounds into their songs.
- The user will be able to choose from: a variety of professionally recorded live instrument samples, synth samples, bass samples, keys samples, pads samples, leads samples, percussion samples, fx samples and more.
- The user will also have the option to create their own sounds or samples using a generated wavetable instrument.
- All instruments, synths, sounds or samples will have parameters that can be adjusted and changed and will have a rack for fx to be added to them within the VST

Screen 1: The “Dashboard”



- Header/Browser: A dropdown or "breadcrumb" menu where the user selects the category
- The Center/Dynamic Interface: This is the "chameleon" part of the program.
 - Case A: User picks a Sample/Live Instrument → Show a waveform display with a few "Character" knobs (e.g., Tone, Air, Reverb etc).
 - Case B: User picks the Wavetable Generator → Show the wavetable oscillator, filter, and ADSR envelopes.
- The Footer (Fixed FX Rack): A horizontal strip at the bottom that is always there, no matter what instrument is loaded (e.g., Delay, Distortion, EQ).



Screen 2 Wavetable Generator option

- When the Wavetable option is chosen the user will be taken to a screen that displays a wave oscillator, following an envelope, filter and or LFO, and Fx rack that its signal travels through.
- The User chooses this option if they want to generate their own sound instead of using a pre recorded sample to start. The screen will look nearly the same as the Sampler screen just with a wave generator in the place of a sample and some slightly different parameters and controls on the wave. The wavetable will appear in the designated box at the top left of the screen and will have controls like octave/pitch adjustment, wavetable position and warp knobs.

Screen 3, Sampler Choice

- When the user selects from the live instrument/sampler categories theyll be taken to the sampler view instead of the wavetable view.
- This Screen will be nearly the same and the sample will go through the envelope, filter, LFO, fx rack etc.
- The Screen will show the same on both options but for the sampler view a sample image will appear in the box at the top left and have some controls like pitch/octave adjustment or sample length.

OSCILLATOR



WAVEFORM: SINE



PITCH



LEVEL



MODE

Browse Categories



Filter



ADSR

SIMPLE FX RACK



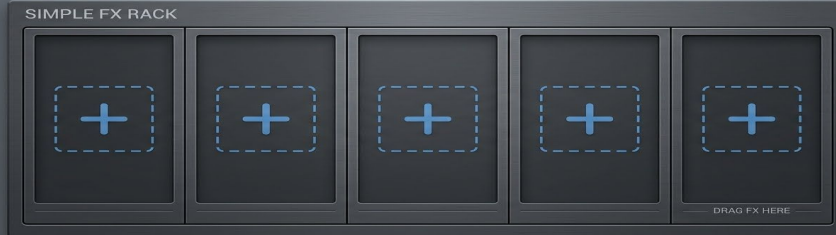
Browse Categories



Filter

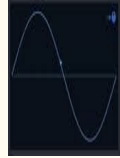


SIMPLE FX RACK





Pseudocode/Logic



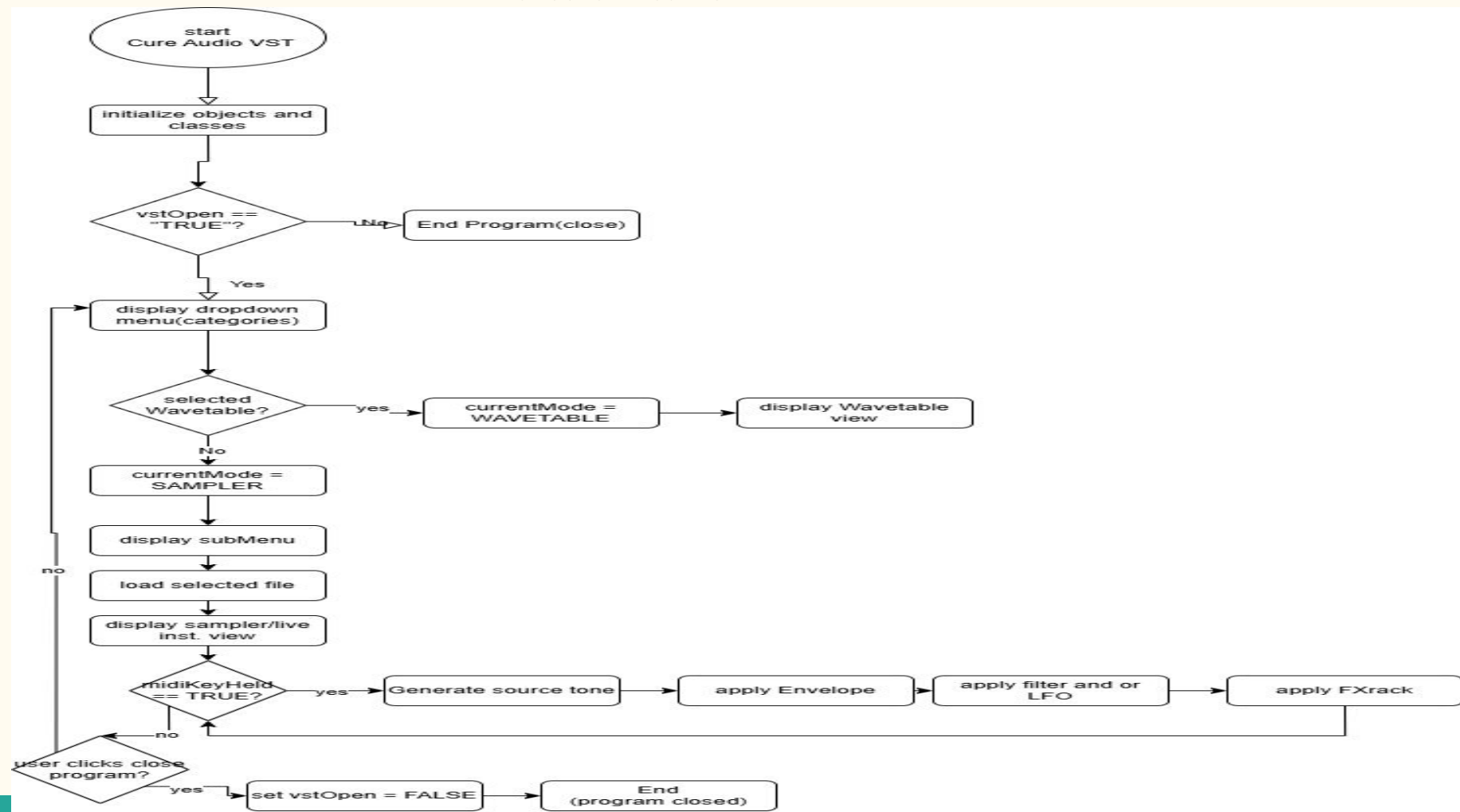
The goal here is 'Seamless Transition'. The user feels like they're using one fluid instrument, though the programming is switching up completely behind the scenes. Here are some key elements that will be worked into the program's logic

- **State Management:** The program must constantly track the CurrentMode. This prevents the "Wavetable" controls from trying to affect a sample by mistake.
- **Dynamic UI Rendering:** Instead of loading two different programs, the VST "hides" or "unhides" specific knob groups (like ADSR or Wavetable Visualizers) based on the chosen instrument.

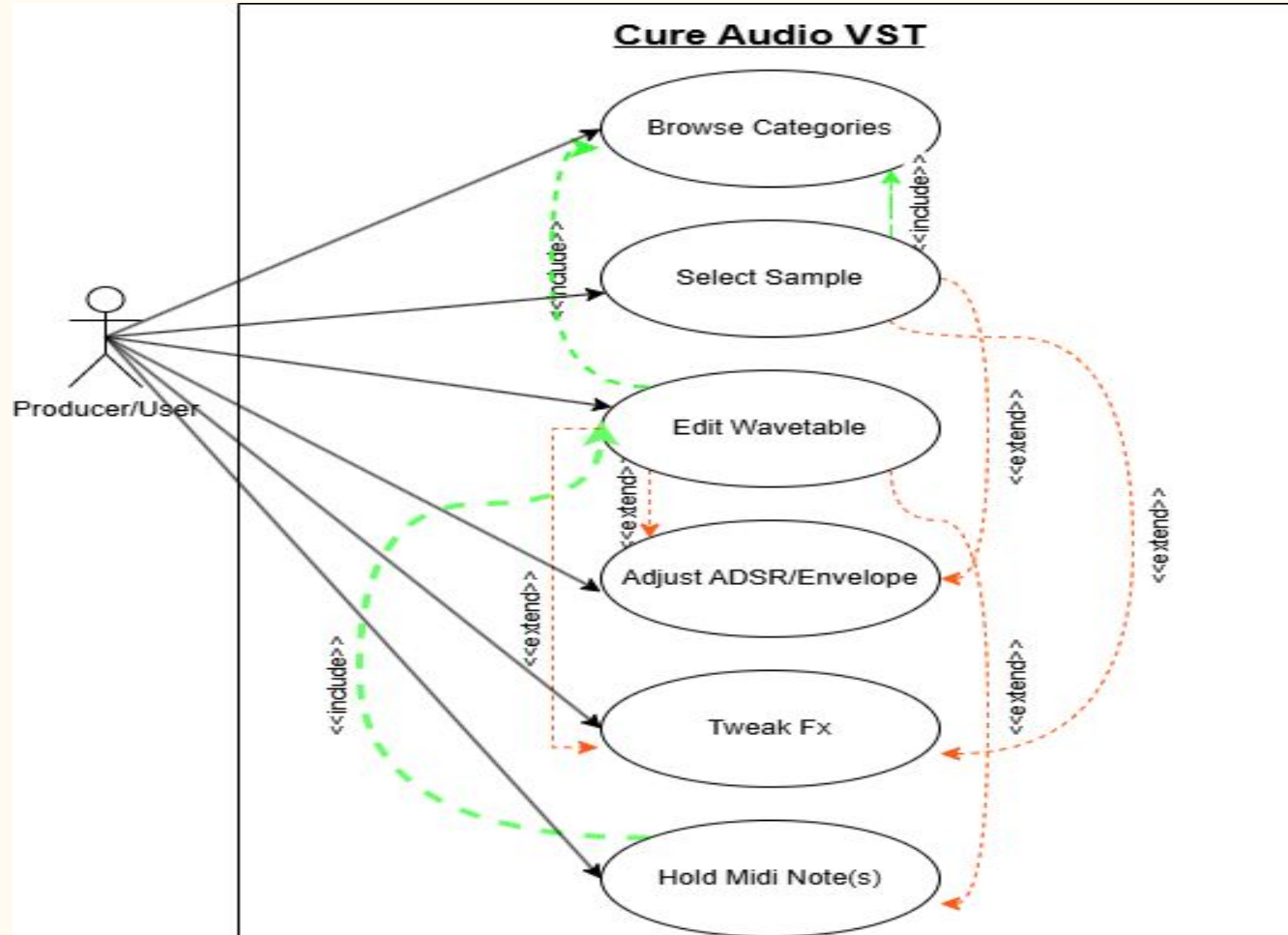
- **Buffer Pre-Loading:** To prevent the DAW from crashing or lagging, the pseudocode needs a logic for "Pre-fetching" samples into RAM (Memory) as soon as the user selects a category.
- **Global FX Persistence:** Explain that the FX_Rack logic is independent of the instrument. Whether it's a drum hit or a synth bass, the signal is routed through the same "Effect Chain" before hitting the output.
- **Resource Cleanup:** When switching from a large Live Instrument (like a multi-sampled Piano) to the Wavetable synth, the program must "Clear Memory" to keep the VST lightweight and fast.



Flowchart



Use Case Diagram



Links for Case Diagrams and Pseudocode

[https://docs.google.com/docume
nt/d/1DTyMwUSmxMG6uC1cI_
_ukBAqAFwW9eC0nrrpgp3_ru
Q/edit?usp=sharing](https://docs.google.com/document/d/1DTyMwUSmxMG6uC1cI_ukBAqAFwW9eC0nrrpgp3_ruQ/edit?usp=sharing)

[https://drive.google.com/file/d/1Org
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S8v/view?usp=drive_link](https://drive.google.com/file/d/1Orgmpjvbgvc4GEoYne9Pk5VWSFrWzS8v/view?usp=drive_link)
